

Question

Quantitative Analysis of Strong Oxidants

NaBiO_3 and PbO_2 are strong oxidants commonly used in industry. Their redox properties can be utilized to conduct volumetric analysis studies.

NaBiO_3 is a light yellow powder that is insoluble in cold water. Accurately weigh 0.8671 g of sodium bismuthate sample, add an appropriate amount of dilute sulfuric acid and MnSO_4 solution to completely convert it to Bi^{3+} , and then titrate with a $\text{H}_2\text{C}_2\text{O}_4$ standard solution with a concentration of $0.1080 \text{ mol} \cdot \text{L}^{-1}$. When the purple-red color of the solution just disappears, 24.89 mL of $\text{H}_2\text{C}_2\text{O}_4$ solution is consumed.

PbO_2 is easily decomposed by heat. Heating a certain amount of PbO_2 yields some solid ($x\text{PbO}_2 \cdot y\text{PbO}$). To investigate the composition of this solid, a certain mass of the solid is taken and added to 30 mL $\text{CH}_3\text{COOH} - \text{CH}_3\text{COONa}$ buffer solution, and then a sufficient amount of KI solid is added. The conical flask is shaken to completely dissolve the solid, resulting in a brown solution. Using starch as an indicator, the solution is titrated with $0.1020 \text{ mol} \cdot \text{L}^{-1}$ of $\text{Na}_2\text{S}_2\text{O}_3$ standard solution until the blue color of the solution just disappears, consuming 24.70 mL of $\text{Na}_2\text{S}_2\text{O}_3$ standard solution. The same mass of the above solid is mixed with dilute nitric acid to obtain a brownish-black powder. The filtrate obtained by filtration is added with a sufficient amount of K_2CrO_4 solution to convert it to PbCrO_4 precipitate. The precipitate is washed and dried, and the mass of the solid obtained is 0.611 g.

Determine the correctness of the following statements

1. The purity of the sodium bismuthate sample (calculated as the mass fraction of NaBiO_3) is 87.69%
2. Determine that $x = 2, y = 3$ through calculation
3. The most accurate chemical equation for the reaction of PbO_2 dissolved in $\text{CH}_3\text{COOH} - \text{CH}_3\text{COONa}$ buffer solution with KI is



A. 1 2 3 are all incorrect.

B. 1 2 3 are all correct

C. Only 1 is correct.

D. Only 2 is correct.

E. Only 3 is correct.

F. Only 1 error

G. Only 2 errors

H. Only 3 errors

Answer

Correct Answer: **D**

Detailed Explanation

Stoichiometric ratio: $5\text{NaBiO}_3 \sim 2\text{MnO}_4^- \sim 5\text{H}_2\text{C}_2\text{O}_4$

CHECKPOINT

1 PTS

The stoichiometric ratio of NaBiO_3 to $\text{H}_2\text{C}_2\text{O}_4$ is 5:5, i.e., 1:1

$$n(\text{NaBiO}_3) = n(\text{H}_2\text{C}_2\text{O}_4) = 0.1080 \times 24.89 \times 10^{-3} = 2.688 \times 10^{-3} \text{ (mol)}$$

$$w(\text{NaBiO}_3) = 2.688 \times 10^{-3} \times 279.97 / 0.8671 = 86.79\%. \text{ Therefore, description 1 is incorrect.}$$

CHECKPOINT

1 PTS

$w(\text{NaBiO}_3) = 86.79\%$, description 1 is incorrect

Stoichiometric ratio: $\text{PbO}_2 \sim \text{I}_2 \sim 2\text{S}_2\text{O}_3^{2-}$

$$n(\text{PbO}_2) = 1/2 \times 0.1020 \times 24.70 \times 10^{-3} = 1.260 \times 10^{-3} \text{ (mol)}$$

CHECKPOINT

1 PTS

Obtained by redox titration: $n(\text{PbO}_2) = 1.260 \times 10^{-3}(\text{ mol})$

Since PbO_2 is insoluble in dilute nitric acid, the gravimetric method corresponds to PbO .

CHECKPOINT

1 PTS

PbO_2 is insoluble in dilute nitric acid

$$n(\text{PbO}) = 0.611/323.2 = 1.89 \times 10^{-3}(\text{ mol})$$

CHECKPOINT

0.5 PTS

$$n(\text{PbO}) = 1.89 \times 10^{-3}(\text{ mol})$$

$$x/y = n(\text{PbO}_2)/n(\text{PbO}) = 1.260 \times 10^{-3}/1.89 \times 10^{-3} = 0.667 \approx 2/3$$

Therefore, $x = 2, y = 3$, description 2 is correct.

CHECKPOINT

1 PTS

$x/y \approx 2/3$, $x = 2, y = 3$, description 2 is correct

Adding sufficient solid KI can complex and dissolve PbI_2 , so it is not reasonable for the product to contain precipitate PbI_2 . Description 3 is incorrect.

CHECKPOINT

1 PTS

It is not reasonable for the product to contain precipitate PbI_2 . Description 3 is incorrect